

TRUNKBELAY · REQUIREMENTS

TrunkBelay requirements

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/trunkbelay		
Source	docs/03-requirements.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Status of every requirement on paper at TRL 3 (TKB-CAL-001); targets unchanged
0.1	2026-09-30	Amish Chadha	Initial scaffold

TrunkBelay requirements

Requirements with their status on paper at TRL 3. The targets are unchanged from version 0.1; they are to be confirmed with the co-design partner. The status column comes from the calculation note TKB-CAL-001.

Requirements that are not met or at risk are open decisions for Amish in the design decisions register (TKB-DEC-001).

Table 1. Requirements

ID	Requirement	Target	Verification (TRL 3 or later)	Status on paper (TRL 3)
R1	Collar holds on a palm trunk	No slip greater than 20 mm under a 2.5 kN (562 lbf) static downward load on wet trunk sections	Proof test on CalRig with cut trunk sections at TRL 3	Met on paper; the friction on wet bark is to be confirmed (locking factor 1.54, settles about 10 mm)
R2	Collar holds when load direction changes	No slip greater than 50 mm when a 100 kg (220 lb) test mass passes the collar during lowering	Drop and transition tests on trunk sections	At risk: the single top chain cannot hold an upward pull; the drill avoids the case (TKB-DEC-001, O1)
R3	Fits common palms	Trunk diameters 200 to 450 mm (8 to 18 in) without tools	Fit trials on surveyed palms	Met: the V and the 110-link chain fit the whole range
R4	Quick to rig	Collar fixed and victim clipped in under 5 minutes by a trained climber	Timed drills at 3 m height	At risk: about 4.5 minutes estimated, a thin margin (TKB-DEC-001, O2)
R5	Controlled lowering	Descent speed held under 0.5 m/s with hand force under 150 N (34 lbf) for a 100 kg (220 lb) load	Lowering tests with test mass	Met on paper: about 80 N at the hand with the brake carabiner
R6	Rope length	Lowers from 25 m (82 ft)	Measured	Met: 30 m of rope, 28 m needed
R7	Portable	Kit mass under 5 kg (11 lb); carried hands-free while climbing	Weighing and climbing trial	Not met: 8.9 kg (TKB-DEC-001, O3)
R8	No damage to the palm	No bark cuts deeper than 5 mm after a full test lowering	Inspection after trials	Met on paper; to confirm by test (pads 1.9 MPa, sleeve 0.6 MPa at 2.5 kN)
R9	Cost	Parts under USD 100 per kit	Costed bill of materials	Not met: USD 758 per kit (TKB-DEC-001, O4)
R10	Teachable	Climbers new to the kit complete a supervised rescue drill after a half-day session	Training trial with partner programme	Cannot be shown on paper; training trial at TRL 4

■ Assumptions

- A second climber is often on site or nearby when a climber is stranded, as in the reported cases.
- The rescuer can reach and work just below the victim using their own climbing device.

- Coconut bark and fibre can carry a collar load without crushing, if the load is spread.
- Victims can be fitted with a sling while hanging head down.

■ Safety

Safety: These requirements describe rope rescue equipment that holds a person at height. Meeting them on paper does not make the kit safe to use. Each is verified by test at TRL 4, and the kit is not used on a person until R1 has been proof-tested on cut trunk sections and the drill has been run with a dummy at low height.